Exercise: Work out the resultant wave from an interference of 2 sound waves with:

i/ Different amplitudes but same wave length and same frequencies.

ii/ Same amplitudes and same wavelengths but different frequencies.

$$s = s_m \cos(kx - \omega t)$$

i/

$$S_1 = S_{1,m} \cos(k_1 x - \omega_1 t)$$
$$S_2 = S_{2,m} \cos(k_2 x - \omega_2 t) \Bigg\} k_1 = k_2$$

$$S_R = S_1 + S_2 \quad \text{if } \omega_1 = \omega_2$$
$$= (S_{1,m} + S_{2,m}) \cos(kx - \omega t)$$

#

ii/

$$S_R = S_1 + S_2 \quad \text{if } \omega_1 \neq \omega_2$$
$$= S_m \left[\cos(kx - \omega_1 t) + \cos(kx - \omega_2 t)\right]$$

$$= 2 S_m \left[\cos\left(\frac{\omega_2 - \omega_1}{2}\right)t\right] \times \left[\cos\left(kx - \frac{(\omega_1 + \omega_2)}{2}t\right)\right]$$

#

3b

① Class Exercise: Young's Experiment

Consider a laser light passing through 2-slits seperated by 5×10^{-3} mm where the fourth bright line on the screen forms at 12.5° with respect to the incident beam. Determine the wavelength of the laser light.

Solution:

[Strategy]

4th bright light

$\Rightarrow 4^{th}$ order constructive interference ($m=4$)

$d = 5\times10^{-3}$ mm

$\theta = 12.5°$

Wavelength formulation:

$d \sin\theta = m\lambda$ [for constructive interference]

Solution :

$$\therefore \lambda = \frac{d \sin\theta}{m}$$

$$= \frac{(5 \times 10^{-3}\ mm) \sin(12.5^\circ)}{4}$$

$$= 2.71 \times 10^{-4}\ mm$$

② Class Exercise: Young's Experiment

Interference patterns have finite number of lines (fringes). Determine the highest-order constructive interference possible with the laser light double-slit setup in the previous problem.

Solution :

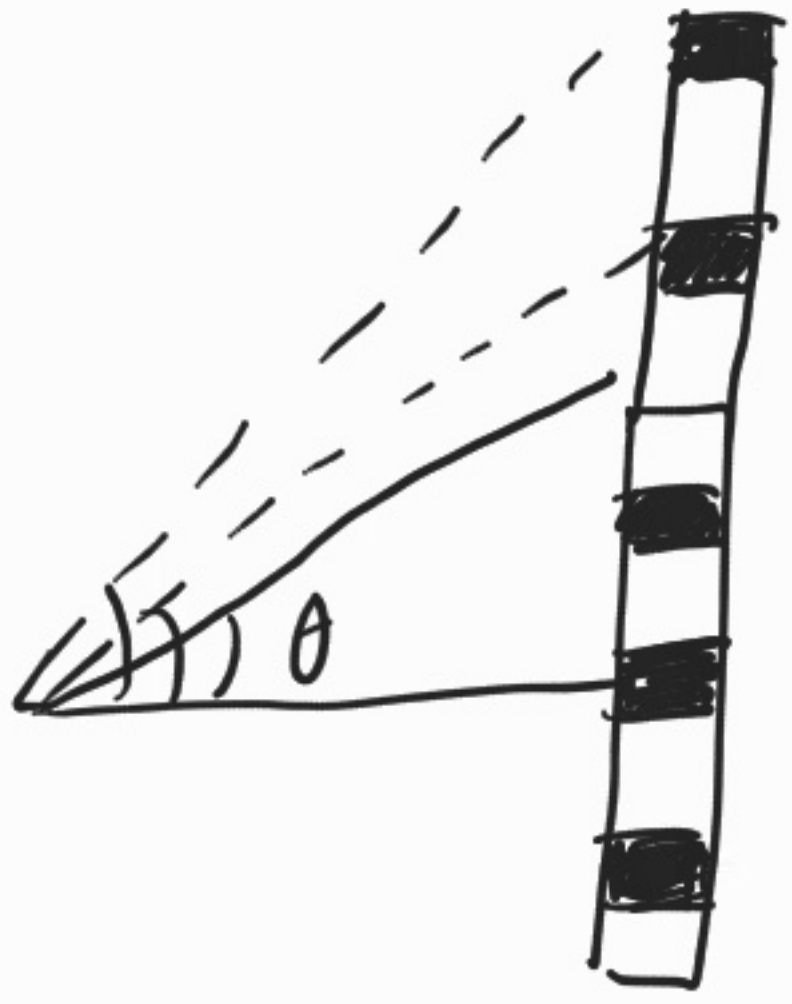
Strategy → $d \sin\theta = m\lambda$

$m = 0, \pm 1, \pm 2, \pm 3, \ldots$

- for the largest value of m, $\sin\theta$ would be at max.

$\therefore \sin\theta = 1$ at $\theta = 90^\circ$

$\theta > 90^\circ$ → doesn't make sense since it reverses the propagation direction.



Solution:

$$\Rightarrow d \sin\theta = m\lambda$$

$$\Rightarrow m = \frac{d \sin\theta}{\lambda} \rightarrow \sin\theta = 1$$

$$\therefore m = \frac{d}{\lambda} = \frac{5\times10^{-3}\ mm}{2.71\times10^{-4}\ mm}$$

$$m = 18.45$$

$\therefore$ largest order for $m = 18$

$\#$